Work along a displacement

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Calculate work and identify when a force does no work.

SUCCESS CRITERIA - I can...

1. Identify the force component along displacement.
2. Use meters and newtons to obtain joules.
3. Distinguish effort from mechanical work on an object.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

01 Warm-up: bring back an earlier idea

A simplified deflection adds **0.3 m/s** sideways for a lead time of **50 s**. **Find the predicted shift** and **name something you** are treating as true.

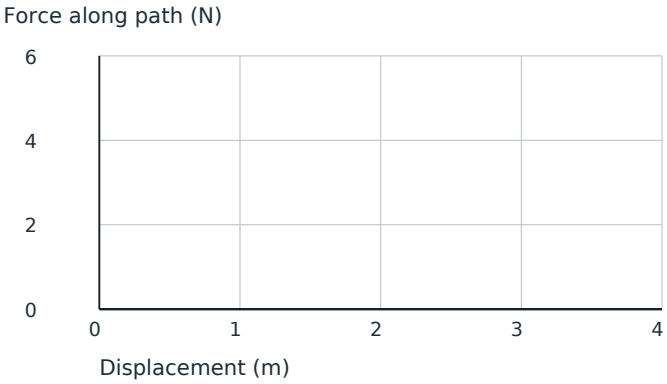
Marzano check - circle: ① ② ③

02 Worked example: explain the prediction

Reference: W = Fd.
Constant force parallel to displacement; W is work. No displacement means no work on the object by that force.

Worked reasoning: A constant **5 N** pull acts along a **3 m** displacement. Work is Fd=**15 J**. The object must move for this force to do mechanical work.

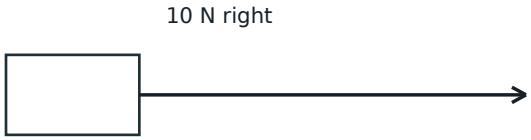
Your turn: 1. **Draw the force** and displacement arrows. 2. **Show the 15 J** as area on a force-versus-position graph.



Marzano check - circle: ① ② ③

03 Read and explain the diagram or equation

A box moves **4 m** right while a **10 N** force acts right. **Draw force** and displacement arrows; **predict the sign of work**.



4 m right

The rectangle is the box. Add the requested force and displacement labels.

Marzano check - circle: ① ② ③

04 Apply the physics

You pull a box with the **10 N** horizontal force shown in question **3**.
1. **Find the work over 4 m**, then over **2 m**.
2. **Draw each force** and displacement; **explain why the shorter trip** changes the work.

Add force and displacement arrows; label the angle.



Force: ____ N Displacement: ____ m

Marzano check - circle: ① ② ③

Work along a displacement

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Continue down each column.

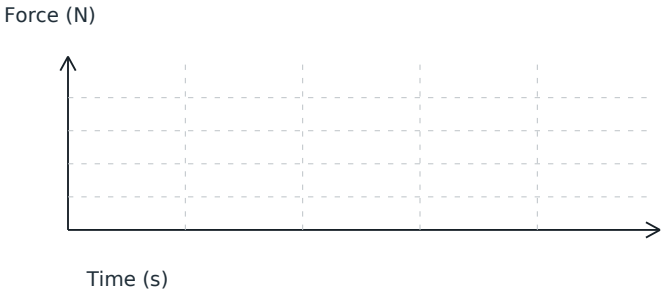
05 Explain your reasoning

A student holds a heavy bag still. Is the upward support force doing mechanical work on the bag during the hold? Explain without denying the student feels tired.

Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

The same object stops from the same speed, but the stopping time is **4** times the original. **Compare average stopping-force magnitudes**.
Show the signed impulse on the force–time graph.



Marzano check - circle: 1 2 3

07 Exit check

A **3 N** force acts forward through **5 m**. **Find its work**.

Add force and displacement arrows; label the angle.

Force: ____ N Displacement: ____ m

Marzano check - circle: 1 2 3

08 STRETCH: test the prediction

1. Design two force-and-distance pairs doing **20 J** of positive work.
2. **Sketch both**.

Add force and displacement arrows; label the angle.

Force: ____ N Displacement: ____ m

Marzano check - circle: 1 2 3